Chapter 65. Fluid Management in the Ventilated Patient

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Figure 65-1

Fluid Management in the Ventilated Patient: Introduction

Fluid management during mechanical ventilation is complicated by both the influence of positive-airway pressure on normal homeostatic control of bodily fluids, and the interaction of mechanical ventilation with fluid status. Hypovolemia may lead to hemodynamic intolerance of positive-airway pressure, and fluid overload may result in both impaired gas exchange and respiratory mechanics and deleterious systemic effects. In patients with acute lung injury (ALI) and the acute respiratory distress syndrome (ARDS), positive fluid balance has been associated with both fewer ventilator-free days, and longer intensive care unit (ICU) stay, and with mortality in prospective randomized[1] and observational[2] studies, respectively.

Physiologic Considerations

Fluid Compartments

See reference 3.

In the normal adult male, total body water accounts for approximately 60% of body weight. In turn, approximately 40% of body weight is intracellular water and approximately 20% is distributed into the extracellular fluid volume, made up of interstitial fluid (approximately 16%), plasma volume (approximately 4%), and usually negligible volumes of lymph and transcellular fluid (cerebrospinal fluid and pericardial, intrapleural, and peritoneal fluid). Tissues such as brain, kidney, liver, and muscle have high water contents (70% to 80%) but adipose tissue has low water content (approximately 10%). Consequently, women, who tend to have more adipose tissue, have a lower total body water (approximately 50% of body weight). Total body water decreases in the elderly because of a loss of muscle mass.

The extracellular volume is distributed in interstitial fluid and plasma volume, and consists of two compartments. Seventy percent of the volume is rapidly equilibrating (approximately 20 minutes), and the remainder slowly equilibrates (approximately 24 hours) in dense connective tissue and bone. Sodium balance regulates the extracellular volume, whereas water balance regulates the intracellular volume.

Water Balance

Water balance is primarily determined by thirst and the renal action of arginine vasopressin, also termed *antidiuretic hormone,* which is secreted from the posterior pituitary following synthesis in the hypothalamus, in response to a wide variety of stimuli, particularly plasma osmolality. Vasopressin activates V_2 receptors on the basolateral surface of the distal renal tubule and collecting duct, leading to an increase in water permeability, and reabsorption of filtrate, through fusion of aquaporin-2 with the luminal membrane. Vasopressin also reduces water clearance by decreasing renal medullary blood flow, and independently increases the renal medullary concentration gradient by stimulating a urea transporter.[4]

Under normal circumstances, a plasma osmolality of 280 mOsm/kg suppresses vasopressin secretion allowing maximal urinary dilution. As osmolality progressively rises to 295 mOsm/kg, so does the secretion of vasopressin, with an associated reduction in free water clearance. The kidney can normally concentrate filtrate up to 1200 mOsm/kg under the influence of vasopressin, although this tends to deteriorate with age and renal dysfunction. Table 65-1 lists other stimuli that influence vasopressin secretion. High-pressure stretch receptors in the aortic arch and carotid sinus sense a significant (>10%) fall in blood pressure (BP), leading to an increase in vasopressin release. As vasopressin also causes vasoconstriction through stimulation of V_1 receptors, this is an important homeostatic response in shock,[4] but appears to be reset within 32 hours of sustained hypovolemia.[5] Stimulation of low pressure stretch receptors in the atria primarily results in an increase in both sympathetic tone and renin, and a decrease in atrial natriuretic peptide (ANP), with vasopressin release unaffected until the systemic BP falls.

Table 65-1: Factors that Influence Vasopressin Secretion

Increase Vasopressin Secretion	Decrease Vasopressin Secretion
Plasma osmolality >280 mOsm/kg	Ethanol
Hypovolemia	Drugs
Hypotension	Narcotic antagonists
High-pressure baroreceptors	Phenytoin
Low-pressure baroreceptors	Clonidine
Angiotensin II	Atrial natriuretic peptide
Pain	
Nausea	
Drugs	
Nicotine	
Narcotics	
Barbiturates	
Carbamazepine	
Amitriptyline	
Cyclophosphamide	
Vincristine	
Clofibrate	
Hypercapnia	
Hypoxemia	

Sodium Balance

The extracellular volume is primarily regulated through control of sodium balance, which is, in turn, regulated through control of effective plasma volume and its composition. The total body sodium content is approximately 4000 mmol; most of which is found extracellularly, and about half is rapidly exchangeable. Although the standard Western diet contains approximately 150 mmol of sodium per 24 hours, this varies widely and urinary sodium excretion varies between 0.2 and 242 mmol per 24 hours,[6] reflecting a balance between sodium input and output. Although a moderate range of total body sodium content is well tolerated, once effective plasma volume is significantly affected, short-term and longer-term homeostatic responses are initiated.

A fall in effective plasma volume leads to activation of baroreceptors with augmentation of myocardial performance and peripheral vascular tone, and defense of plasma volume through shift of fluid from the interstitium. Longer-term responses include reduced sodium loss by the kidney and sweat glands, through a direct effect of aldosterone. When the baroreceptors are stimulated the increase in sympathetic tone reduces sodium loss through reduced glomerular filtration rate, and through increased tubular sodium reabsorption, both through a direct effect and through the actions of increased renin, angiotensin II, and aldosterone. In addition, ANP is released from the cardiac atria in response to stretch, and directly increases glomerular filtration rate through afferent arteriolar vasodilation, increases renal medullary blood flow, antagonizes vasoconstriction secondary to angiotensin II, and decreases sodium reabsorption by the collecting duct. Dopamine is produced in the kidney following conversion from L-dopa under the action of the cytosolic enzyme L-amino acid decarboxylase present in the proximal tubules.[7] This is upregulated following a high-salt diet, leading to increased urinary sodium loss as dopamine acts to inhibit sodium reabsorption in the proximal tubule,[8] and contributes to the increase in urine output sometimes seen following administration of low-dose dopamine. The renal synthesis of prostaglandins, such as prostaglandin E_2 and prostacyclin (PGI_2), tends to maintain renal blood flow and glomerular filtration rate through vasodilation, and directly increase water and sodium excretion. Consequently, in stressed patients cyclooxygenase inhibitors can precipitate renal dysfunction. Dopaminergic renal vasodilation in part acts through release of PGI_2, because administration of dopaminergic antagonists leads to reduced urinary prostaglandins and loss of dopaminergic vasodilation,[9] perhaps explaining why low-dose dopamine appears to be ineffective in septic ICU patients[10] who already have a prostaglandin-driven kidney.

Influence of Positive-Pressure Ventilation on Fluid Balance

Positive-pressure ventilation and positive end-expiratory pressure (PEEP) raise intrathoracic pressure, resulting in reduced venous return and transmural pressure (see Chapter 36), with consequent complex neurohumoral responses leading to sodium and water retention. Because assisted, supported, and spontaneous modes of ventilation progressively ameliorate the elevation of intrathoracic pressure and its consequences, different ventilator modes variably reduce venous return. Reductions in stroke volume, cardiac output, and BP then lead to stimulation of high-pressure baroreceptors, and altered regional blood flow. Both low- and high-pressure baroreceptor stimulation lead to increased sympathetic outflow, and release of renin, aldosterone, and ANP. Renal denervation does not prevent sodium and water retention.[11] Angiotensin-converting enzyme inhibitors[12] and deliberate hypervolemia,[11] however, reduce sodium and water retention during positive-pressure ventilation.

Right atrial transmural pressure and stretch are also reduced by PEEP and positive-pressure ventilation, and this leads to reduced secretion of ANP,[13,14] with consequent reduction in water and sodium excretion reversed by restoration of venous return with lower body positive pressure. PEEP levels above 10 cm H_2O may lead to an increase in central venous pressure (CVP), and regional venous pressures, which in the kidney contribute to reduced sodium and water excretion, independent of neurohumoral effects.[15]

In summary, various neurohumoral responses to positive-pressure ventilation lead to retention of sodium and water, as a homeostatic response to raised intrathoracic pressure. A major consequence of this response is expanded plasma volume, and a tendency toward systemic and pulmonary edema.

The Starling Equation

The major difference between plasma volume and interstitial fluid is the lower concentration of plasma proteins in the interstitial fluid, typically 40% of their plasma concentration. Although this concentration difference has little effect on the osmotic pressure between these two compartments, it leads to an important difference in oncotic pressure, and 80% of this is attributed to differences in albumin concentration. The normal plasma osmotic pressure is approximately 5500 to 6000 mm Hg, with approximately 28 mm Hg contributed by plasma proteins, despite having an osmolality of approximately 1.2 mOsm/kg.

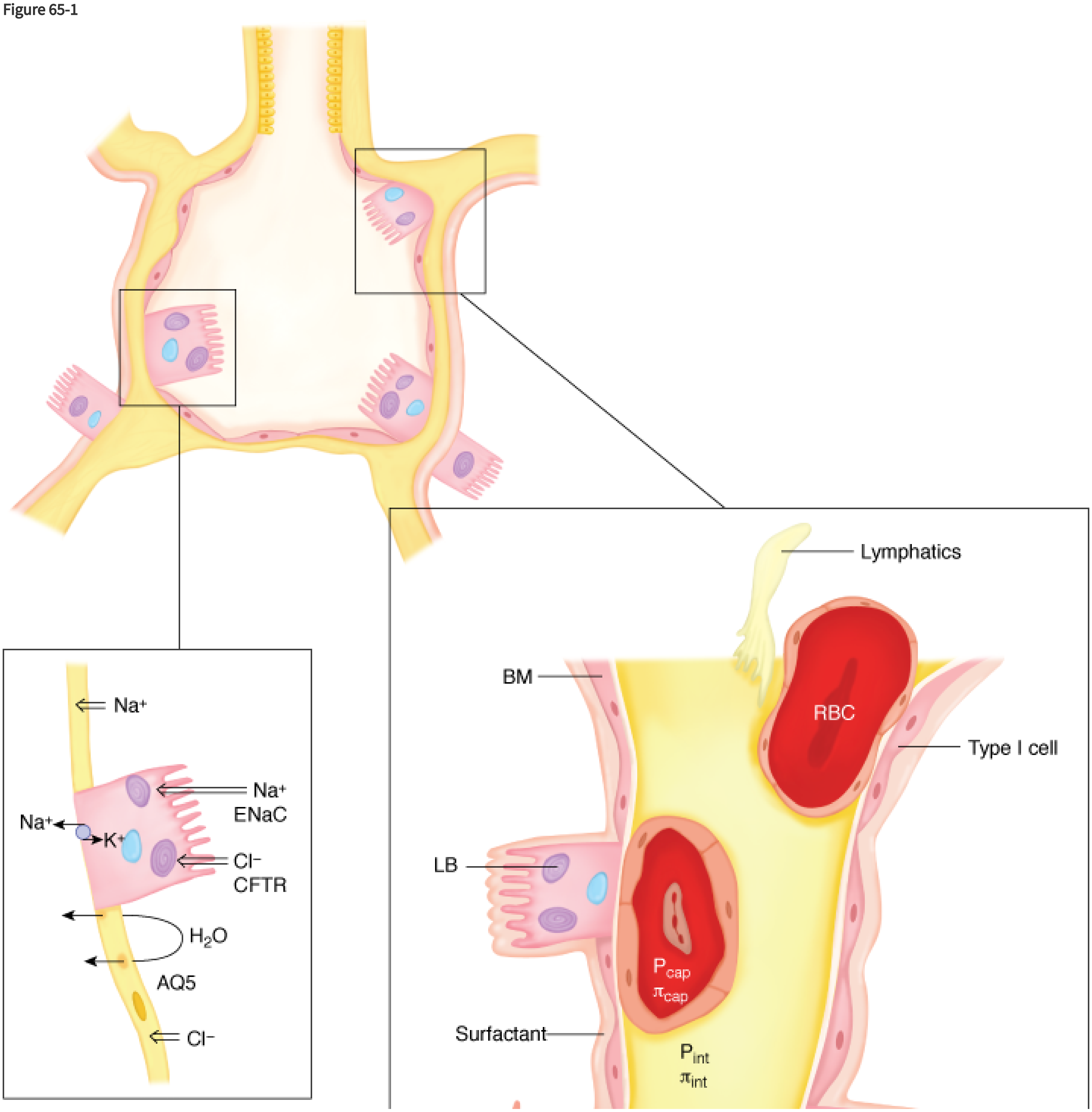
The Starling equation quantitates the transvascular flux of fluids across the microcirculation (J_V). It is usually written as:

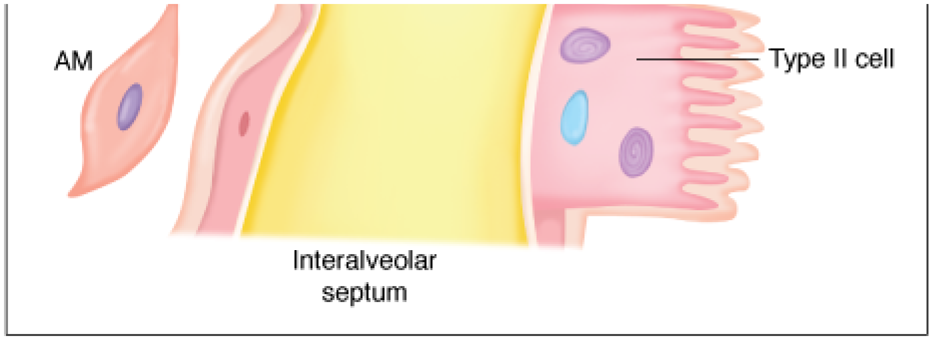
$$J_V = K_{f,c}\left([P_{cap} - P_{int}] - \sigma\,[\pi_{cap} - \pi_{int}]\right)$$

where $K_{f,c}$ is the capillary filtration coefficient, P_{cap} is the hydrostatic microvascular pressure, P_{int} is the interstitial pressure, π_{cap} is the plasma oncotic pressure, π_{int} is the interstitial oncotic pressure, and σ is the osmotic reflection coefficient. $K_{f,c}$ is determined by both endothelial hydraulic conductance and endothelial surface area, and σ is a measure of protein selectivity. The osmotic reflection coefficient is thought to be 1 in the cerebral microcirculation where the blood–brain barrier effectively prevents protein flux, and approximately 0.7 to 0.8 in the normal pulmonary microcirculation, although this is markedly reduced during lung injury.[14] In a typical systemic microcirculatory bed, the arterial end of P_{cap} is 30 mm Hg and the venous end is 10 mm Hg. Assuming σ equals 1, P_{int} is −3 mm Hg, π_{cap} is 28 mm Hg, and π_{int} is 8 mm Hg, the net driving pressure out of the capillary at the arterial end of the microcirculation will be ([30−3] − [28−8]) or 13 mm Hg, although the effective π_{int} may be a little lower.[16] At the venous end of the microcirculation the net driving pressure into the capillary will be 7 mm Hg, and most of the filtered fluid is reabsorbed, with the lymphatics draining the remainder.

In the healthy lung, P_{cap} is usually assumed to be 7 mm Hg, P_{int} as −8 mm Hg, and π_{int} as 14 mm Hg. Consequently, the driving pressure across the pulmonary circulation is thought to be positive (Fig. 65-1), leading to net filtration of fluid, with lymphatic absorption usually estimated to be approximately 20 mL/hour. The final filtration rate is determined by the capillary surface area, convective forces, and diffusive forces. When P_{cap} is suddenly raised, the filtration rate may increase more than threefold.[16]

Figure 65-1





P_{cap} +7 mm Hg P_{int} −8 mm Hg

π_{cap} +28 mm Hg π_{int} +14 mm Hg

$$\Delta P = [7 - -8] - [28 - 14] = +1 \text{ mm Hg}$$

Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Schematic of an alveolus, associated interstitium, and capillary network in the normal lung. Typical pressures involved in fluid flux result in net positive filtration of fluid. The epithelium is both a tight barrier with pore size approximately one-tenth the endothelium, and participates in vectorial ion and water and movement out of the alveolus. Sodium is absorbed at the epithelial surface through a sodium channel (*ENaC*) and actively moved across the basolateral membrane into the interstitium by Na^+,K^+-adenosine triphosphatase (ATPase). *AM,* alveolar macrophage; *AQ5,* aquaporin 5; *BM,* basement membrane; *CFTR,* cystic fibrosis transmembrane conductance regulator; *LB,* lamellar body; P_{cap}, pulmonary capillary pressure; π_{int}, interstitial oncotic pressure; π_{cap}, pulmonary capillary oncotic pressure; P_{int}, interstitial pressure; *RBC,* red blood cell.

There are a number of safety factors that are thought to help prevent alveolar edema. These include low epithelial permeability (pore size approximately 10% that of the endothelium), low alveolar surface tension reflecting normal surfactant function, active vectorial transport of ions across the epithelium (see Fig. 65-1), and favorable interstitial function and lymphatic drainage (edema fluid moves along a negative-pressure gradient centrally away from the alveolus, while an associated reduction in the negative pressure and dilution of interstitial oncotic pressure reduce filtration).[17] Changes in either Starling forces or these safety factors can lead to pulmonary edema. For example, an increase in P_{cap} is the basis for hydrostatic pulmonary edema; a decrease in P_{int}, as might be seen with an obstructed airway and vigorous respiratory efforts, is thought to cause postobstructive pulmonary edema; although a decrease in σ is the basis for permeability pulmonary edema, concurrent surfactant dysfunction leads to an increase in surface tension and a decrease in P_{int}, also favoring the development of edema.[18–20] Remodeling of the lung parenchyma in chronic heart failure and conditions of persistently elevated P_{cap} such as mitral stenosis allow patients to tolerate a relatively high P_{cap} without developing marked pulmonary edema. Models of chronic heart failure suggest important homeostatic responses including a reduction in $K_{f,c}$, which together with further reduction in surface tension associated with increased surfactant content, can prevent pulmonary edema despite elevated P_{cap}.[21,22]

Fluid Targets

In formulating an approach to fluid management in the ventilated patient, a balance between parsimonious and generous fluid therapy needs to be considered. Although this is commonly termed the “dry or wet” approach, and this may be an appropriate general description for particular groups of patients, most clinicians classify fluid therapy as (a) maintenance, (b) replacement, and (c) resuscitation fluids. In general, the tendency of ventilated patients to retain fluids, and the benefits of fluid restriction, argue for the dry approach provided there is due attention to adequate resuscitation. A consensus statement classified fluid restriction as grade IIa evidence in ALI and ARDS.[23]

Maintenance Fluids

The volume of appropriate maintenance fluids in a ventilated patient is usually the most contentious of these parameters. As noted in Table 65-2, approximately 1150 mL of fluid per day should be adequate maintenance in ventilated patients; this amount is less than in nonventilated subjects because all gases are humidified. In practice, metabolic production of water is usually ignored, allowing a total intake of 1500 mL/day; this represents a baseline volume that needs to be reviewed following clinical and biochemical assessment of plasma volume and total body water. Nevertheless, in balance, this volume should be sufficient to generate a urine output of approximately 800 mL/day or 0.5 mL/kg/hour; because the normal daily solute load excreted by the kidney is approximately 600 mOsm, this is easily achieved by the normal kidney with this urine volume by concentrating urine to 700 to 800 mOsm/L. The planned sum of enteral and parenteral maintenance fluids, however, may be less than this because of obligatory fluids given with drug infusions and hydrostatic pressure transducer flush (approximately 3 mL/hour per transducer). Other factors influencing maintenance fluids include the size of the patient, covert losses, such as a diaphoresis, fever that increases fluid loss by approximately 10 mL/kg/day for each degree of temperature elevation, and the ease of supplying adequate nutrition. Although 1500 mL is given as an adequate maintenance fluid volume, many centers use greater volumes, and many patients tolerate greater maintenance fluid volumes.

Table 65-2: Typical Fluid Balance in Healthy (Nonventilated) Subjects and Ventilated Patients

	Healthy Subjects (mL)	Ventilated Patients (mL)
Typical obligatory fluid (water) losses		
Gastrointestinal fluid	200	200
Insensible skin loss	500	500
Humidification of inhaled air	500	0
Urine output	1000	800
Total	2200	1500
Typical fluid (water) intake		
Metabolically generated water	350	350
Water content of food	750	0
Remaining fluid intake	1100	1150
Total	2200	1500

In practice, many clinicians ignore the metabolic production of water, and prescribe 1500 mL of fluid in a ventilated patient.

The Dry Approach

The neurohumoral response to positive-pressure ventilation tends to retain sodium and water, which may lead to both peripheral and pulmonary edema. Daily weights are inconvenient and infrequently performed in ventilated patients. Peripheral edema needs to be carefully sought, and is often evident in the limbs of bedridden patients, and as a wedge-shaped swelling in the flanks. Pulmonary edema is typically detected as dependent crackles on auscultation, but this is a late sign and requires an appropriately placed stethoscope. Other techniques include chest radiographs and measurement of extravascular lung water. In addition to lack of sensitivity, these methods may be troublesome to interpret in disease states such as ALI.

Moderate hypohydration (mean 4.5% loss in body weight) leads to a reversible improvement in lung volume and airflow resistance in normal subjects.[24] Excess fluids may lead to pulmonary edema, which is associated with impaired oxygenation, prolonged ventilation and ICU stay, and difficulty weaning. In both ALI and ARDS,[25] however, and in acute cardiogenic pulmonary edema,[26] extravascular lung water does not correlate with CVP or pulmonary artery occlusion pressure (PAOP). Although the PAOP is an important determinant of the pulmonary capillary filtration pressure, the extravascular lung water is also influenced by permeability and temporal effects. In acute pulmonary edema, empiric treatment with diuretics, nitrates, and ventilator support will often lead to marked reduction of CVP and PAOP before resolution of the pulmonary edema.[26] Indeed, there may be transient hypovolemia secondary to extravasation of fluid in the lung, requiring volume loading.

In ALI and ARDS, extravascular lung water is usually elevated despite normal filtration pressure. Excess lung water portends a worse outcome.[27] Patients with ARDS who achieve a significant reduction in PAOP[28] or total body water, as estimated from weight loss or cumulative fluid balance,[29] are more likely to survive. Prospective, randomized studies report improved oxygenation in hypoproteinemic patients with ALI when a negative fluid balance was produced using furosemide and concentrated albumin;[30] and management of pulmonary edema according to lung water, as compared to PAOP, can lead to a lower cumulative fluid balance, fewer ventilator days, and shorter ICU stay.[31] Prevention of acute left-ventricular failure by diuresis allows weaning in some ventilator-dependent patients with chronic obstructive pulmonary disease.[32] Positive fluid balance is associated with increased mortality in both ALI,[2] and in critically ill patients with acute kidney injury.[33] Liberal fluid management in ALI reduces ventilator-free days and increases ICU stay,[1] with a trend to increased dialysis (14% vs. 10%, $p = 0.06$) consistent with reduced recovery of renal function with fluid overload in acute kidney injury.[33] An important aspect of these data is that the patients were enrolled following the diagnosis of ALI in ventilated patients, and may not apply to the initial resuscitation phase. Taken together, these data suggest that accumulation of excess fluid is common, often difficult to detect, and may be associated with serious adverse events, and mortality.

Replacement Fluids

There is little argument regarding replacement of excess fluid loss. Once the volume becomes significant or contributes to difficulties with fluid balance, it should be replaced. Table 65-3 lists typical electrolyte compositions of gastrointestinal fluids. In the setting of a postobstructive diuresis, it is common to use 0.45% normal saline with 10 mmol KCl per 500-mL flask; however, greater certainty regarding composition of excess fluid losses can be gained by measuring the electrolyte composition.

Table 65-3: Composition of Gastrointestinal Losses

	Volume (mL)	Na^+ (mmol/L)	K^+ (mmol/L)	Cl^- (mmol/L)	HCO_3^- (mmol/L)
Salivary	500 to 1000	50	20	40	30
Gastric	1500	60 to 100	10	150	30(H^+)
Pancreatic	400 to 1000	140	5	75	115
Bile	400 to 1000	140	5	100	35 to 60
Ileal	1000 to 3000	140	5	70 to 115	30 to 50
Colon		60	70	15	30

As noted above, gastric fluid is usually acidic with an H^+ concentration of approximately approximately 30 mmol/L; this will be markedly reduced by the use of proton pump inhibitors or H_2-antagonists.

Resuscitation Fluids

Critically ill ventilated patients commonly require administration of resuscitation fluids to correct hypovolemia. Underpinning the use of resuscitation fluids is the relationship between increased ventricular preload and increased stroke volume. The trigger for a bolus of fluids often appears fairly obvious: for example, hypotension in a patient with overt bleeding. On other occasions, it may be unclear as to whether a patient will be fluid responsive or whether hemodynamic support is indicated. Yet, early recognition of the need for fluid resuscitation with rapid and titrated administration cannot be underestimated.

Choice of Resuscitation Fluids

A variety of isotonic and hypertonic fluids have been examined for resuscitation, with varied results. This is often simplified as the "crystalloid versus colloid" controversy. Colloids are fluids with oncotically active contents, such as albumin (molecular weight 66 kDa), which theoretically should ensure that the fluid primarily expands the plasma volume, with little contribution to interstitial volume and edema. If permeability is increased, however, these oncotic substances may also leak across into the interstitium possibly contributing to edema formation. Table 65-4 lists typical electrolyte compositions of commonly used crystalloids and colloids.

Table 65-4: Composition of Typical Crystalloids and Colloids

	Osmolality	Na^+	K^+	Cl^-	Organic Anion
0.9% saline[a]	300	150		150	
Hartmann's solution[a]	274	129	5	109	29 (lactate)
Ringer's lactate	272	130	4	109	28 (lactate)
Plasma-Lyte 148	296	140	5	98	50 (acetate, gluconate)
Albumex 4[b]	260	140		128	6.4 (octanoate)
Gelofusine[c]	283	144		120	
Dextran 40 in 0.9% saline	325	150		150	
Dextran 70 in 0.9% saline	306	150		150	
Hetastarch 6% in 0.9% saline	310	154		154	
7.5% saline in 6% dextran 70	2567	1283		1283	

[a]Baxter Healthcare, Toongabbie, Australia.

[b]CSL Limited, Parkville, Australia.

[c]Braun, Bella Vista, Australia.

Blood and Blood Products

These fluids preferentially expand the plasma volume, and may be indicated for appropriate correction of coagulopathy or maintenance of oxygen carriage and delivery. An important distinction needs to be made between stable and unstable patients when considering transfusion. Data from a large randomized study of stable critically ill patients supports use of a restrictive transfusion policy using a hemoglobin threshold of 7 g/dL, to aim for a target hemoglobin of 7 to 9 g/dL.[34] Although even mild degrees of anemia can result in angina in patients with severe coronary artery disease, observational data of transfusion practice in acute myocardial infarction do not strongly argue for a higher transfusion threshold.[35] In unstable patients a different approach should be considered, and recent recommendations for management of severe sepsis and septic shock suggest a transfusion threshold of 100 g/L during the initial 6-hour resuscitation period.[36]

Crystalloid or Colloid Resuscitation

Comparing colloids and crystalloids for fluid resuscitation in critically ill patients, a recent Cochrane review[37] and a megatrial (total: 6997 patients) of saline versus albumin (SAFE)[38] found no difference in outcome. In the SAFE study more than 60% of the patients were ventilated, and the trigger for fluid loading was based on common clinical grounds and clinician judgment. Although specific groups, such as post–cardiac surgery, post–liver or post–kidney transplantation, and burns were excluded, the data from the SAFE study are widely applicable to critically ill patients.[39] In current practice,[40] there is wide variation in the use of crystalloids and colloids between countries, but colloids are used more often. Crystalloids offer equivalent efficacy, cost, and fewer adverse effects, while colloids offer more rapid volume correction, longer duration of action, reduced risk of pulmonary and interstitial edema, and they may be more appropriate for the clinical setting. Although clinical practice will be influenced by the SAFE study, it seems likely that issues such as patient subgroups, cost and availability, and clinician preference will still influence choice of fluid for volume expansion.

Albumin

Albumin is usually administered as either an isoosmotic solution (e.g., 4% albumin, which has 40 g/L albumin and sodium 140 mmol/L) or as a more concentrated form (e.g., 20% albumin). It appears to be the safest of the commonly used colloids[41] with a reported serious adverse event rate of approximately 1 per 10^6 infusions.[42] Compared to albumin, the anaphylactoid reaction rates of comparable colloids are higher with risk ratios approximately 2 for dextran, approximately 4.5 for hydroxyethyl starch, and approximately 12 for gelatin, with a pooled rate of approximately 9 per 10^5 infusions.[41] In normal subjects, albumin has a metabolic half-life of 16 to 20 days, with a turnover of 12 to 15 g/day.[43] The normal escape of albumin from plasma occurs at about 10% per hour, which is markedly increased during sepsis with 32% of the initial rise following albumin administration lost by 4 hours.[43] Increased catabolism and reduced production of albumin by the liver also contribute to reduced serum albumin levels in critically ill patients. Albumin binds numerous substances such as fatty acids, calcium, thyroxine, amino acids, and hydrogen ions; in addition, many commonly used drugs such as warfarin, phenytoin, midazolam, and antibiotics are highly protein bound.[23] Consequently, hypoalbuminemia may have important physiologic and pharmacologic effects.

Post hoc analysis of the SAFE study suggested that patients with severe traumatic brain injury may have a higher mortality rate with albumin resuscitation (relative risk [RR]: 1.88; 95% confidence interval [CI]: 1.31 to 2.70; $p < 0.001$),[44] and that patients with severe sepsis[45] may have a reduced mortality rate following multivariate logistic regression with adjustment for baseline factors, adjusted odds ratio for death for albumin versus saline was 0.71 (95% CI: 0.52 to 0.97; $p = 0.03$). In patients with severe liver disease and spontaneous bacterial peritonitis, albumin reduces renal dysfunction and mortality.[46] Circulatory dysfunction following paracentesis in cirrhotic patients, defined as an increase in plasma renin activity, is reduced with albumin as compared to gelatin, dextran,[47] or saline,[48] and albumin appears preferable to gelatin for reversal of diuretic-induced hepatic encephalopathy, possibly caused by a reduction in oxidant stress.[49] Finally, albumin and furosemide may have some benefit in hypoproteinemic patients with ALI.[30]

Hydroxyethyl Starch

Hydroxyethyl starch is available in a variety of molecular weights (high, medium, and low: 450 to 480, 130 to 200, and 40 to 70 kDa), C2:C6 ratios (high and low: >8, <8), and molar substitutions (high or low: 0.6 to 0.7, 0.4 to 0.5), which alter their breakdown, intravascular half-life, and in vivo molecular weight.[50] There are, however, significant concerns regarding their safety. The bleeding risk after cardiac surgery may be increased, and starches have been associated with hepatic dysfunction, pruritus, and renal dysfunction;[51] the high in vivo molecular weight[50] is thought to contribute to hyperoncotic renal injury.[52] The newer, smaller, lower half-life starches and volumes less than 1500 mL tend to be associated with few adverse effects,[50] and the starches are commonly used as colloids for resuscitation. There are concerns, however, that even the newer starches increase the risk of renal dysfunction,[53] and their routine use has not been recommended.[54]

Other Colloids

Dextrans are rarely used as volume expanders in ventilated patients because of increases in both bleeding risk and allergic reaction,[41] and possible increased risk of renal dysfunction.[54] They are glucose polymers of either average molecular weight 40 or 70 kDa, with dextran 70 sometimes used to reduce red cell and platelet sludging. Gelatins are a group of volume expanders with a low molecular weight (35 kDa), leading to a half-life of approximately 2 hours. Again allergic reactions are more common, and the main use of gelatins is outside the ICU or for short-lived periods of volume expansion, such as post–cardiac surgery.

Crystalloids

Crystalloids are usually isotonic solutions (e.g., 0.9% saline), but hypertonic solutions (e.g., 7.5% saline) have been increasingly used, particularly in the prehospital management of trauma where this may correct hypotension and reduce cerebral edema. A double-blind, prospective, randomized study, however, comparing 7.5% saline with Ringer lactate solution in 229 prehospital trauma patients, with a systolic BP less than 100 mm Hg and a Glasgow Coma Scale score less than 9, found similar rates for hospital mortality and 6-month neurologic outcome.[55] Large volumes of 0.9% saline, or colloids that contain this electrolyte composition, may lead to a non–anion gap acidosis. An alternative approach is to use balanced crystalloids, such as Ringer lactate (or Hartmann solution), which contain a modest amount of organic anion (e.g., lactate or acetate) that is metabolized to bicarbonate, and tend to maintain a more normal acid–base state. Various adverse effects, such as increased production of nitric oxide, ALI, hypotension, renal dysfunction, impaired gastric perfusion, nausea, abdominal pain, and bleeding have been attributed to this acidosis.[56,57] Apart from the change in acid–base state, however, there is little evidence of improved clinical outcome with a balanced solution.[58]

Isotonic fluids freely distribute into the extracellular space leading to edema and greater volume requirement than colloids. Although the ratio of crystalloid to colloid needed to achieve the same effect is expected to be at least 3:1, in the SAFE study it was 1.4:1.[38] This was associated with a small but significantly greater heart rate, however, and both lower CVP and transfused volume in the crystalloid arm. In normal subjects, rapid infusion of 0.9% saline at 30 mL/kg over 20 to 30 minutes may reduce forced vital capacity, reduce forced expiratory volume in 1 second,[59] and produce premature airway closure and hypoxemia.[60] Maximum oxygen consumption may also be reduced, possibly because of edema of skeletal muscle and impaired O_2 diffusion.[61] A smaller volume of rapidly infused saline (10 mL/kg), however, only leads to adverse effects in patients with left-ventricular dysfunction.[61] Although there are no direct comparisons of crystalloid with colloid loading, these adverse effects, and the benefits of reducing lung water, argue for care with crystalloids. If a choice is available, it also seems sensible to use a balanced electrolyte fluid, provided there is adequate hepatic function and perfusion to metabolize the associated organic anion.

Monitoring Fluid Therapy

An excess of body water in relation to sodium is manifest as hyponatremia, and deficiency of water in relation to sodium as hypernatremia, which is often associated with a disproportionate increase of urea relative to plasma creatinine. Factitious results (e.g., hyponatremia in the setting of hyperglycemia) and coexistent disease processes, such as Addison disease, may result in hyponatremia, while sepsis and gastrointestinal blood are other common causes of a disproportionate rise in urea.

Monitoring of both the trigger and response to volume loading can be very useful because excessive fluid therapy can result in adverse events, and only 40% to 72% of critically ill patients increase stroke volume or cardiac output with volume loading.[61] The SAFE study[38] used simple triggers for volume loading, while recent guidelines for management of severe sepsis and septic shock[36] used a lower mean BP threshold (65 mm Hg) and aimed for central or mixed venous oxygen saturation equal to or greater than 70% based on improved survival and reduced organ dysfunction in a randomized study of early goal-directed therapy.[63] Although perioperative mortality may be reduced in high-risk patients with a strategy that targeted increased oxygen delivery,[64] studies in critically ill patients show no benefit[65] or a worse outcome,[66] perhaps reflecting later intervention when organ dysfunction has already been initiated.

Minimally Invasive Methods

Chest Radiographs

In addition to heart size, and the presence of pulmonary infiltrates and pleural effusions, the vascular pedicle width, measured as the horizontal distance between a line dropped from the point where the left subclavian artery leaves the aortic arch to the point where the right main bronchus crosses the superior vena cava, may provide an additional useful measure of volume status. A vascular pedicle width greater than 63 to 70 mm can help distinguish hydrostatic from permeability pulmonary edema,[67] and it falls with negative fluid balance in ALI.[68] A decrease in vascular pedicle width of 5 mm, however, corresponded to a 3.2-L negative balance,[68] suggesting that this technique may not be sensitive enough for acute fluid management decisions, and most patients with permeability pulmonary edema have some signs usually ascribed to volume overload.[67]

Dynamic Measures

Blood Pressure Variation

Positive-pressure ventilation alters left-ventricular loading conditions leading to both systolic pressure and pulse pressure, and stroke-volume variation during the respiratory cycle, which is predictive of an increase in cardiac output, following a fluid bolus.[69] Validation, however, has mostly been in stable patients without impaired left-ventricular function, and there are important preconditions and caveats. There should be absence of inspiratory or expiratory effort and a stable cardiac rhythm. Because these dynamic measures have also not been shown to be predictive of fluid responsiveness when used with smaller tidal volumes or in the presence of pulmonary hypertension,[69] and respiratory effort is present in many critically ill patients, they have limited applicability. Similar reservations apply to the recently described use of respiratory variations in the preejection period (the time between the Q wave on the electrocardiogram and the upstroke of the arterial pressure waveform)[70] and stroke volume[71] as a measure of fluid responsiveness.

Passive leg raising to 45 degrees results in variable autotransfusion and a temporary increase in venous return, which has been used to test fluid responsiveness. Although the subsequent change in pulse pressure is predictive of fluid responsiveness, the change in cardiac output has a significantly higher predictive value.[72] If patients who increased their CVP by at least 2 mm Hg with passive leg raising are analyzed,[73] then both measures have a higher predictive performance, but again the change in pulse pressure is inferior. An increase in CVP was present in about half of the subjects studied, of whom one-third were fluid responsive, suggesting limited clinical utility.

Ultrasound

Although left-ventricular end-diastolic area is not predictive, echocardiography has been used to assess respiratory variation in the diameter of the vena cava as a measure of fluid responsiveness. In septic patients, a threshold of 36% for superior vena cava collapse during inspiration is both sensitive and specific,[74] and is not correlated with the CVP, but is similar to dynamic pulse pressure changes as a measure of fluid responsiveness. Echocardiography, however, offers the advantage of detection of severe right-ventricular failure, which may lead to false-positive results with pulse pressure variation. Respiratory changes in inferior vena cava diameter also appear promising, but may be invalidated by raised intraabdominal pressure.[75] Because both techniques require similar conditions to those needed for BP variation measures, they may not be widely applicable in the ICU. Lung ultrasound has been promulgated as a measure of subpleural interstitial pulmonary edema,[76] which may reflect PAOP when permeability is normal and in the absence of rapid changes in filtration pressure.

Parameters derived from the esophageal Doppler, such as the heart-rate corrected time to peak flow (a preload measure) and the descending aortic flow, may be used to predict volume responsiveness. A number of studies have used the esophageal Doppler to guide perioperative fluid management; fewer complications and shorter hospital stays were associated with greater fluid administration.[77] The use of this technique, however, appears limited in most ventilated ICU patients; they will not tolerate an esophageal probe without additional sedation. A fixed 70% of the aortic blood flow is assumed to pass to the descending aorta, and patients with irregular cardiac rhythms, an intraaortic balloon pump, or esophageal disease are usually excluded. Although good agreement of esophageal Doppler with thermodilution cardiac output has been reported,[78] others have found substantial variability.[79,80]

Invasive Measures

Central Venous Catheter

The CVP is determined by the venous return and right-heart function. It is increased by many factors, including high levels of PEEP and decreased venous capacitance; it does not correlate particularly well with right or left end-diastolic volume in critically ill patients. Although a number of studies have not found the CVP predictive of volume responsiveness,[62] it may still be a useful measure.[81] Above a CVP of 12 mm Hg, few patients are volume responsive, and changes in the CVP may be helpful. An unchanged CVP after volume loading is suggestive of volume responsiveness, whereas a large increase is not. Similarly, the absence of a fall in CVP with spontaneous respiratory effort suggests lack of volume response.[81] Rivers et al[63] used a target CVP of 8 to 12 mm Hg, and continuously measured the central venous oxygen saturation in their goal-directed group with a target of equal to or greater than 70%. Under normal conditions, a mixed venous oxygen saturation of 75% corresponds to an intracellular partial pressure of oxygen (P_{O_2}) of 11 mm Hg, but this falls to 0.8 mm Hg at a saturation of 50%. Consequently, central venous access may be useful in determining fluid responsiveness or the need for other means of augmenting cardiac output.

Transpulmonary thermodilution is a technique that allows intermittent and continuous measurement of cardiac output, and closely agrees with pulmonary artery thermodilution measurement.[82] Additional measures include extravascular lung water and global end-diastolic volume, which is a measure of the total volume of the four heart chambers, and may be a useful preload measure.[83] However, concern has been expressed about the accuracy, including thermally silent areas as a result of hypoxic pulmonary vasoconstriction, of the extravascular lung water measure.[84,85] In addition to central venous access, this technique requires insertion of a thermodilution arterial catheter, which is usually inserted via the femoral artery although it can be inserted via the brachial artery. Although it is a promising technique, it does increase the risk of complications, and requires further investigation.

Pulmonary Artery Catheter

Although the PAOP should be a measure of left-ventricular preload, it correlates poorly with left-ventricular end-diastolic volume in critically ill patients, and is a poor marker of volume responsiveness. Measurement of the PAOP requires insertion of a pulmonary artery catheter, which may allow measurement of a number of variables, including cardiac output, pulmonary artery pressure, mixed venous oxygen saturation, and right-ventricular volumes; numerous derived variables can be subsequently calculated. The accuracy and interpretation, however, of some of these data have been questioned, and serious complications from the catheter and its insertion are well described. Retrospective analysis found that use of the pulmonary artery catheter in critically ill patients was associated with increased mortality, hospital stay, and costs despite careful adjustment for severity.[86] A more recent prospective observational study found that there was a marked increase in postoperative events.[87] Nevertheless, data measured by the pulmonary artery catheter can be extremely useful in particular patients, and central venous oxygen saturation,[63] a surrogate for mixed venous oxygen saturation, may be an important end point for resuscitation.

Mortality, morbidity, and complications are not influenced by the presence of a pulmonary artery catheter in patients with shock and/or ARDS.[88,89] Although high-risk surgical patients may have a higher rate of pulmonary embolism,[90] this was not confirmed by the ARDS Network trial comparing pulmonary artery with central venous catheter guided management in ALI.[89] Apart from more insertion-related arrhythmias with the pulmonary artery catheter, this large trial did not confirm the suggestion of more complications. In the subgroup of surgical patients, randomization to the pulmonary artery catheter group was associated with increased fluid administration and fewer ventilator-free days,[91] suggesting that interpretation and consequent management decisions have a greater influence on outcome than does the catheter itself.

Conclusion

As both physiologic measures and outcome are influenced by fluid management, an integrated approach is essential to the management of ventilated patients. The tendency to retain sodium and water, the adverse effects of pulmonary and peripheral edema, and improved outcomes from both observational and randomized clinical trials argue for a parsimonious approach to maintenance fluids, provided there is adequate resuscitation. Although colloids appear as safe as crystalloids, there may be particular circumstances in which one is preferable, even though it remains unclear whether balanced salt solutions improve clinical outcomes. Dynamic measures of fluid responsiveness, such as respiratory variations in BP, are superior to static measures, such as an isolated CVP or PAOP reading, but common clinical conditions often render these data invalid. Perhaps, more important than the choice of fluid, or assessment of fluid responsiveness, is the definition of integrated pathways that can be simply applied to patient care.

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